

■ 3.6.5 Converting Power Series to Normal Expressions

`Normal[expr]` convert a power series to a normal expression.

Converting power series to normal expressions.

As discussed above, power series in *Mathematica* are represented in a special internal form, which keeps track of such attributes as their expansion order.

For some purposes, you may want to convert power series to normal expressions. From a mathematical point of view, this corresponds to truncating the power series, and assuming that all higher order terms are zero.

This generates a power series, with four terms.

```
In[1]:= t = Series[ Exp[x], {x, 0, 4} ]
```

$$\text{Out}[1]= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + O[x]^5$$

Squaring the power series gives you another power series, with the appropriate number of terms.

```
In[2]:= t^2
```

$$\text{Out}[2]= 1 + 2 x + 2 x^2 + \frac{4 x^3}{3} + \frac{2 x^4}{3} + O[x]^5$$

`Normal` truncates the power series, giving a normal expression.

```
In[3]:= Normal[%]
```

$$\text{Out}[3]= 1 + 2 x + 2 x^2 + \frac{4 x^3}{3} + \frac{2 x^4}{3}$$

You can now apply standard algebraic operations.

```
In[4]:= Factor[%]
```

$$\text{Out}[4]= \frac{3 + 6 x + 6 x^2 + 4 x^3 + 2 x^4}{3}$$